

3

BIFURCATIONS

3.0 Introduction

As we've seen in Chapter 2, the dynamics of vector fields on the line is very limited: all solutions either settle down to equilibrium or head out to $\pm\infty$. Given the triviality of the dynamics, what's interesting about one-dimensional systems? Answer: *Dependence on parameters*. The qualitative structure of the flow can change as parameters are varied. In particular, fixed points can be created or destroyed, or their stability can change. These qualitative changes in the dynamics are called **bifurcations**, and the parameter values at which they occur are called **bifurcation points**.

Bifurcations are important scientifically—they provide models of transitions and instabilities as some *control parameter* is varied. For example, consider the buckling of a beam. If a small weight is placed on top of the beam in Figure 3.0.1, the beam can support the load and remain vertical. But if the load is too heavy, the vertical position becomes unstable, and the beam may buckle.

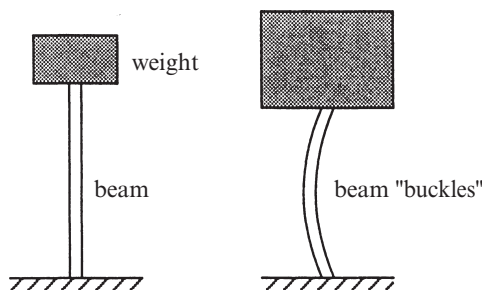


Figure 3.0.1

Here the weight plays the role of the control parameter, and the deflection of the beam from vertical plays the role of the dynamical variable x .

One of the main goals of this book is to help you develop a solid and practical understanding of bifurcations. This chapter introduces the simplest examples: bifurcations of fixed points for flows on the line. We'll use these bifurcations to model such dramatic phenomena as the onset of coherent radiation in a laser and the outbreak of an insect population. (In later chapters, when we step up to two- and three-dimensional phase spaces, we'll explore additional types of bifurcations and their scientific applications.)

We begin with the most fundamental bifurcation of all.

3.1 Saddle-Node Bifurcation

The saddle-node bifurcation is the basic mechanism by which fixed points are *created and destroyed*. As a parameter is varied, two fixed points move toward each other, collide, and mutually annihilate.

The prototypical example of a saddle-node bifurcation is given by the first-order system

$$\dot{x} = r + x^2 \quad (1)$$

where r is a parameter, which may be positive, negative, or zero. When r is negative, there are two fixed points, one stable and one unstable (Figure 3.1.1a).

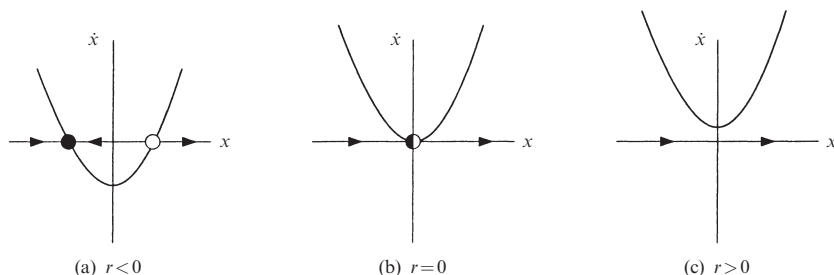


Figure 3.1.1

As r approaches 0 from below, the parabola moves up and the two fixed points move toward each other. When $r = 0$, the fixed points coalesce into a half-stable fixed point at $x^* = 0$ (Figure 3.1.1b). This type of fixed point is extremely delicate—it vanishes as soon as $r > 0$, and now there are no fixed points at all (Figure 3.1.1c).

In this example, we say that a *bifurcation* occurred at $r = 0$, since the vector fields for $r < 0$ and $r > 0$ are qualitatively different.

Graphical Conventions

There are several other ways to depict a saddle-node bifurcation. We can show a stack of vector fields for discrete values of r (Figure 3.1.2).

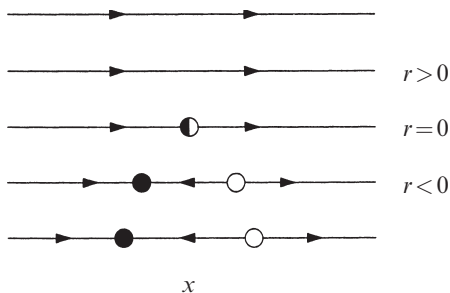


Figure 3.1.2

This representation emphasizes the dependence of the fixed points on r . In the limit of a *continuous* stack of vector fields, we have a picture like Figure 3.1.3. The curve shown is $r = -x^2$, i.e., $\dot{x} = 0$, which gives the fixed points for different r . To distinguish between stable and unstable fixed points, we use a solid line for stable points and a broken line for unstable ones.

However, the most common way to depict the bifurcation is to invert the axes of Figure 3.1.3. The rationale is that r plays the role of an independent variable,

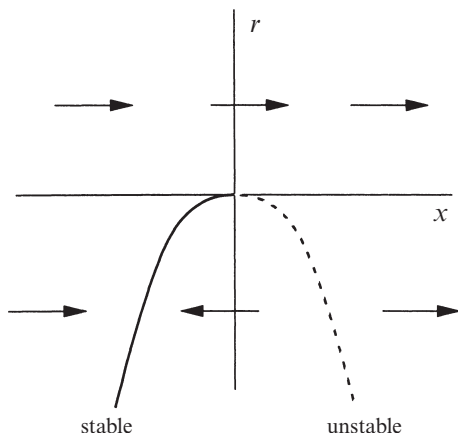


Figure 3.1.3

and so should be plotted horizontally (Figure 3.1.4). The drawback is that now the x -axis has to be plotted vertically, which looks strange at first. Arrows are sometimes included in the picture, but not always. This picture is called the **bifurcation diagram** for the saddle-node bifurcation.

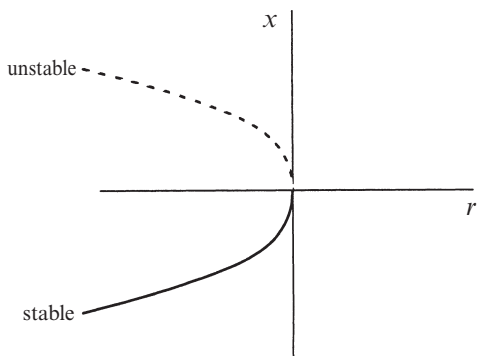


Figure 3.1.4

Terminology

Bifurcation theory is rife with conflicting terminology. The subject really hasn't settled down yet, and different people use different words for the same thing. For example, the saddle-node bifurcation is sometimes called a *fold bifurcation* (because the curve in Figure 3.1.4

has a fold in it) or a *turning-point bifurcation* (because the point $(x, r) = (0, 0)$ is a “turning point.”) Admittedly, the term *saddle-node* doesn’t make much sense for vector fields on the line. The name derives from a completely analogous bifurcation seen in a higher-dimensional context, such as vector fields on the plane, where fixed points known as saddles and nodes can collide and annihilate (see Section 8.1).

The prize for most inventive terminology must go to Abraham and Shaw (1988), who write of a *blue sky bifurcation*. This term comes from viewing a saddle-node bifurcation in the other direction: a pair of fixed points appears “out of the clear blue sky” as a parameter is varied. For example, the vector field

$$\dot{x} = r - x^2 \quad (2)$$

has no fixed points for $r < 0$, but then one materializes when $r = 0$ and splits into two when $r > 0$ (Figure 3.1.5). Incidentally, this example also explains why we use the word “bifurcation”: it means “splitting into two branches.”

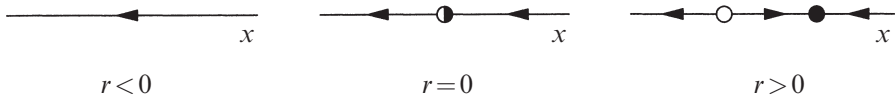


Figure 3.1.5

EXAMPLE 3.1.1:

Give a linear stability analysis of the fixed points in Figure 3.1.5.

Solution: The fixed points for $\dot{x} = f(x) = r - x^2$ are given by $x^* = \pm\sqrt{r}$. There are two fixed points for $r > 0$, and none for $r < 0$. To determine linear stability, we compute $f'(x^*) = -2x^*$. Thus $x^* = +\sqrt{r}$ is stable, since $f'(x^*) < 0$. Similarly $x^* = -\sqrt{r}$ is unstable. At the bifurcation point $r = 0$, we find $f'(x^*) = 0$; the linearization vanishes when the fixed points coalesce. ■

EXAMPLE 3.1.2:

Show that the first-order system $\dot{x} = r - x - e^{-x}$ undergoes a saddle-node bifurcation as r is varied, and find the value of r at the bifurcation point.

Solution: The fixed points satisfy $f(x) = r - x - e^{-x} = 0$. But now we run into a difficulty—in contrast to Example 3.1.1, we can’t find the fixed points explicitly as a function of r . Instead we adopt a geometric approach. One method would be to graph the function $f(x) = r - x - e^{-x}$ for different values of r , look for its roots x^* , and then sketch the vector field on the x -axis. This method is fine, but there’s an easier way. The point is that the two functions $r - x$ and e^{-x} have much more familiar graphs than their difference $r - x - e^{-x}$. So we plot $r - x$ and e^{-x} on the same

picture (Figure 3.1.6a). Where the line $r - x$ intersects the curve e^{-x} , we have $r - x = e^{-x}$ and so $f(x) = 0$. Thus, intersections of the line and the curve correspond to fixed points for the system. This picture also allows us to read off the direction of flow on the x -axis: the flow is to the right where the line lies above the curve, since $r - x > e^{-x}$ and therefore $\dot{x} > 0$. Hence, the fixed point on the right is stable, and the one on the left is unstable.

Now imagine we start decreasing the parameter r . The line $r - x$ slides down and the fixed points approach each other. At some critical value $r = r_c$, the line becomes *tangent* to the curve and the fixed points coalesce in a saddle-node bifurcation (Figure 3.1.6b). For r below this critical value, the line lies below the curve and there are no fixed points (Figure 3.1.6c).

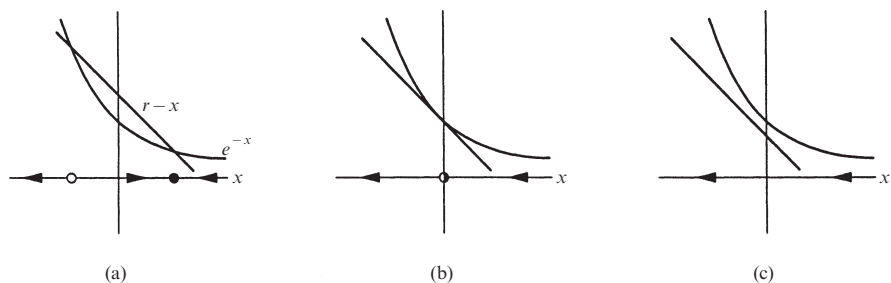


Figure 3.1.6

To find the bifurcation point r_c , we impose the condition that the graphs of $r - x$ and e^{-x} intersect *tangentially*. Thus we demand equality of the functions *and* their derivatives:

$$e^{-x} = r - x$$

and

$$\frac{d}{dx} e^{-x} = \frac{d}{dx} (r - x).$$

The second equation implies $-e^{-x} = -1$, so $x = 0$. Then the first equation yields $r = 1$. Hence the bifurcation point is $r_c = 1$, and the bifurcation occurs at $x = 0$. ■

Normal Forms

In a certain sense, the examples $\dot{x} = r - x^2$ or $\dot{x} = r + x^2$ are representative of *all* saddle-node bifurcations; that's why we called them "prototypical." The idea is that, close to a saddle-node bifurcation, the dynamics typically look like $\dot{x} = r - x^2$ or $\dot{x} = r + x^2$.

For instance, consider Example 3.1.2 near the bifurcation at $x = 0$ and $r = 1$. Using the Taylor expansion for e^{-x} about $x = 0$, we find

$$\begin{aligned}\dot{x} &= r - x - e^{-x} \\ &= r - x - \left[1 - x + \frac{x^2}{2!} + \dots \right] \\ &= (r-1) - \frac{x^2}{2} + \dots\end{aligned}$$

to leading order in x . This has the same algebraic form as $\dot{x} = r - x^2$, and can be made to agree exactly by appropriate rescalings of x and r .

It's easy to understand why saddle-node bifurcations typically have this algebraic form. We just ask ourselves: how can two fixed points of $\dot{x} = f(x)$ collide and disappear as a parameter r is varied? Graphically, fixed points occur where the graph of $f(x)$ intersects the x -axis. For a saddle-node bifurcation to be possible, we need two nearby roots of $f(x)$; this means $f(x)$ must look locally “bowl-shaped” or parabolic (Figure 3.1.7).

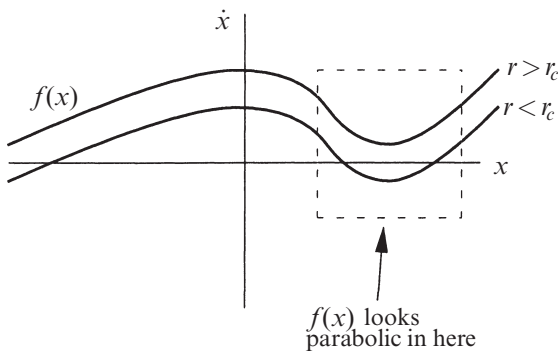


Figure 3.1.7

Now we use a microscope to zoom in on the behavior near the bifurcation. As r varies, we see a parabola intersecting the x -axis, then becoming tangent to it, and then failing to intersect. This is exactly the scenario in the prototypical Figure 3.1.1.

Here's a more algebraic version of the same argument. We regard f as a function of both x and r , and examine the behavior of $\dot{x} = f(x, r)$ near the bifurcation at $x = x^*$ and $r = r_c$. Taylor's expansion yields

$$\begin{aligned}\dot{x} &= f(x, r) \\ &= f(x^*, r_c) + (x - x^*) \frac{\partial f}{\partial x} \Big|_{(x^*, r_c)} + (r - r_c) \frac{\partial f}{\partial r} \Big|_{(x^*, r_c)} + \frac{1}{2} (x - x^*)^2 \frac{\partial^2 f}{\partial x^2} \Big|_{(x^*, r_c)} + \dots\end{aligned}$$

where we have neglected quadratic terms in $(r - r_c)$ and cubic terms in $(x - x^*)$. Two of the terms in this equation vanish: $f(x^*, r_c) = 0$ since x^* is a fixed point, and $\partial f / \partial x|_{(x^*, r_c)} = 0$ by the tangency condition of a saddle-node bifurcation. Thus

$$\dot{x} = a(r - r_c) + b(x - x^*)^2 + \dots \quad (3)$$

where $a = \partial f / \partial x|_{(x^*, r_c)}$ and $b = \frac{1}{2} \partial^2 f / \partial x^2|_{(x^*, r_c)}$. Equation (3) agrees with the form of our prototypical examples. (We are assuming that $a, b \neq 0$, which is the typical case; for instance, it would be a very special situation if the second derivative $\partial^2 f / \partial x^2$ also happened to vanish at the fixed point.)

What we have been calling prototypical examples are more conventionally known as **normal forms** for the saddle-node bifurcation. There is much, much more to normal forms than we have indicated here. We will be seeing their importance throughout this book. For a more detailed and precise discussion, see Guckenheimer and Holmes (1983) or Wiggins (1990).

3.2 Transcritical Bifurcation

There are certain scientific situations where a fixed point must exist for all values of a parameter and can never be destroyed. For example, in the logistic equation and other simple models for the growth of a single species, there is a fixed point at zero population, regardless of the value of the growth rate. However, such a fixed point may *change its stability* as the parameter is varied. The transcritical bifurcation is the standard mechanism for such changes in stability.

The normal form for a transcritical bifurcation is

$$\dot{x} = rx - x^2. \quad (1)$$

This looks like the logistic equation of Section 2.3, but now we allow x and r to be either positive or negative.

Figure 3.2.1 shows the vector field as r varies. Note that there is a fixed point at $x^* = 0$ for *all* values of r .

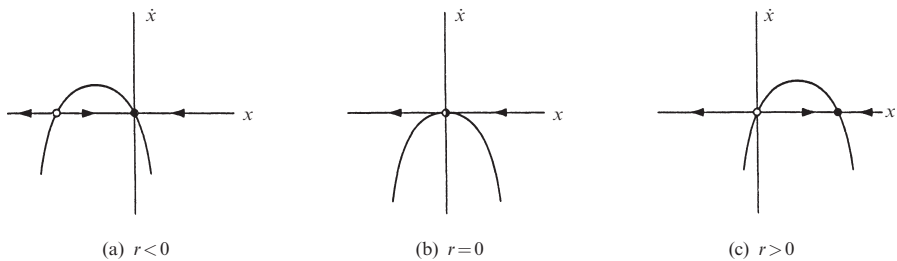


Figure 3.2.1

For $r < 0$, there is an unstable fixed point at $x^* = r$ and a stable fixed point at $x^* = 0$. As r increases, the unstable fixed point approaches the origin, and coalesces with it when $r = 0$. Finally, when $r > 0$, the origin has become unstable, and $x^* = r$ is now stable. Some people say that an *exchange of stabilities* has taken place between the two fixed points.

Please note the important difference between the saddle-node and transcritical bifurcations: in the transcritical case, the two fixed points don't disappear after the bifurcation—instead they just switch their stability.

Figure 3.2.2 shows the bifurcation diagram for the transcritical bifurcation. As in Figure 3.1.4, the parameter r is regarded as the independent variable, and the fixed points $x^* = 0$ and $x^* = r$ are shown as dependent variables.

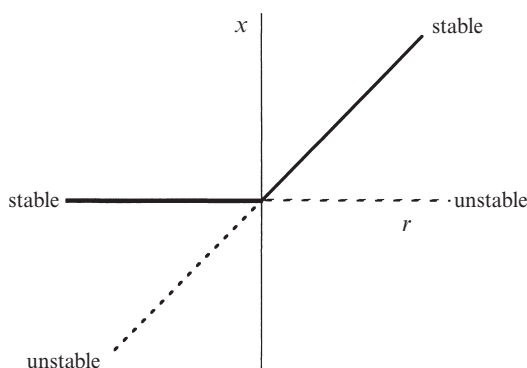


Figure 3.2.2

EXAMPLE 3.2.1:

Show that the first-order system $\dot{x} = x(1 - x^2) - a(1 - e^{-bx})$ undergoes a transcritical bifurcation at $x = 0$ when the parameters a, b satisfy a certain equation, to be determined. (This equation defines a *bifurcation curve* in the (a, b) parameter space.) Then find an approximate formula for the fixed point that bifurcates from $x = 0$, assuming that the parameters are close to the bifurcation curve.

Solution: Note that $x = 0$ is a fixed point for all (a, b) . This makes it plausible that the fixed point will bifurcate transcritically, if it bifurcates at all. For small x , we find

$$\begin{aligned} 1 - e^{-bx} &= 1 - \left[1 - bx + \frac{1}{2}b^2x^2 + O(x^3) \right] \\ &= bx - \frac{1}{2}b^2x^2 + O(x^3) \end{aligned}$$

and so

$$\begin{aligned} \dot{x} &= x - a(bx - \frac{1}{2}b^2x^2) + O(x^3) \\ &= (1 - ab)x + (\frac{1}{2}ab^2)x^2 + O(x^3). \end{aligned}$$

Hence a transcritical bifurcation occurs when $ab = 1$; this is the equation for the bifurcation curve. The nonzero fixed point is given by the solution of $1 - ab + (\frac{1}{2}ab^2)x \approx 0$, i.e.,

$$x^* \approx \frac{2(ab-1)}{ab^2}.$$

This formula is approximately correct only if x^* is small, since our series expansions are based on the assumption of small x . Thus the formula holds only when ab is close to 1, which means that the parameters must be close to the bifurcation curve. ■

EXAMPLE 3.2.2:

Analyze the dynamics of $\dot{x} = r \ln x + x - 1$ near $x = 1$, and show that the system undergoes a transcritical bifurcation at a certain value of r . Then find new variables X and R such that the system reduces to the approximate normal form $\dot{X} \approx RX - X^2$ near the bifurcation.

Solution: First note that $x = 1$ is a fixed point for all values of r . Since we are interested in the dynamics near this fixed point, we introduce a new variable $u = x - 1$, where u is small. Then

$$\begin{aligned}\dot{u} &= \dot{x} \\ &= r \ln(1+u) + u \\ &= r \left[u - \frac{1}{2}u^2 + O(u^3) \right] + u \\ &\approx (r+1)u - \frac{1}{2}ru^2 + O(u^3).\end{aligned}$$

Hence a transcritical bifurcation occurs at $r_c = -1$.

To put this equation into normal form, we first need to get rid of the coefficient of u^2 . Let $u = av$, where a will be chosen later. Then the equation for v is

$$\dot{v} = (r+1)v - (\frac{1}{2}ra)v^2 + O(v^3).$$

So if we choose $a = 2/r$, the equation becomes

$$\dot{v} = (r+1)v - v^2 + O(v^3).$$

Now if we let $R = r + 1$ and $X = v$, we have achieved the approximate normal form $\dot{X} \approx RX - X^2$, where cubic terms of order $O(X^3)$ have been neglected. In terms of the original variables, $X = v = u/a = \frac{1}{2}r(x-1)$. ■

To be a bit more accurate, the theory of normal forms assures us that we can find a change of variables such that the system becomes $\dot{X} = RX - X^2$, with

strict, rather than approximate, equality. Our solution above gives an approximation to the necessary change of variables. For careful treatments of normal form theory, see the books of Guckenheimer and Holmes (1983), Wiggins (1990), or Manneville (1990).

3.3 Laser Threshold

Now it's time to apply our mathematics to a scientific example. We analyze an extremely simplified model for a laser, following the treatment given by Haken (1983).

Physical Background

We are going to consider a particular type of laser known as a solid-state laser, which consists of a collection of special “laser-active” atoms embedded in a solid-state matrix, bounded by partially reflecting mirrors at either end. An external energy source is used to excite or “pump” the atoms out of their ground states (Figure 3.3.1).

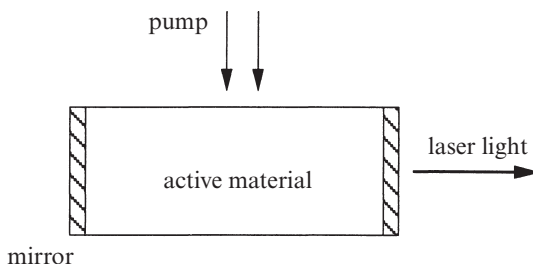


Figure 3.3.1

Each atom can be thought of as a little antenna radiating energy. When the pumping is relatively weak, the laser acts just like an ordinary *lamp*: the excited atoms oscillate independently of one another and emit randomly phased light waves.

Now suppose we increase the strength of the pumping. At first nothing different happens, but then suddenly, when the pump strength exceeds a certain threshold, the atoms begin to oscillate in phase—the lamp has turned into a ***laser***. Now the trillions of little antennas act like one giant antenna and produce a beam of radiation that is much more coherent and intense than that produced below the laser threshold.

This sudden onset of coherence is amazing, considering that the atoms are being excited completely at random by the pump! Hence the process is *self-organizing*: the coherence develops because of a cooperative interaction among the atoms themselves.